

AI Based Heart Attack Detection Using ECG Signal Analysis

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Abstract—Cardiovascular diseases are a leading cause of death worldwide, with heart attacks being among the most critical medical emergencies. Early detection of abnormal heart activity can significantly reduce complications and improve survival rates. The electrocardiogram (ECG) is a widely used non invasive technique for monitoring the electrical activity of the heart; however, traditional ECG interpretation relies on manual analysis, which is time consuming and not suitable for continuous monitoring. This paper presents an Artificial Intelligence based heart attack detection system using ECG signal analysis. The proposed system performs signal preprocessing to remove noise, extracts important features such as heart rate, RR intervals, and waveform patterns, and uses a machine learning model to classify ECG signals as normal or abnormal for early identification of potential heart attack risk. The system aims to support healthcare professionals in decision making, reduce dependency on manual interpretation, and enable continuous monitoring through wearable or portable devices, ultimately contributing to early diagnosis, preventive care, and improved patient safety.

Keywords— Artificial Intelligence, Electrocardiogram (ECG), Heart Attack Detection, Machine Learning, Cardiac Monitoring, Healthcare Technology.

I. INTRODUCTION

Cardiovascular diseases have emerged as one of the leading causes of mortality worldwide, placing a significant burden on global healthcare systems. Among these conditions, heart attacks are particularly lifethreatening because they often occur suddenly and require immediate medical intervention. A heart attack typically occurs when the blood supply to a portion of the heart muscle is partially or completely blocked, most commonly due to the accumulation of fatty deposits in the coronary arteries. If not detected and treated promptly, this condition can lead to irreversible cardiac damage or death. In recent years, the incidence of heart attacks has increased not only among the elderly but also among younger populations. This trend is especially alarming in developing countries such as India, where lifestyle changes, rising stress levels, unhealthy dietary habits, and reduced physical activity have contributed to the growing prevalence of heart-related disorders.

Early detection of cardiac abnormalities plays a crucial role in minimizing severe complications and improving patient outcomes. Continuous monitoring of heart activity enables the identification of warning signs before the onset of a major cardiac event. The electrocardiogram (ECG) is one of the most widely used noninvasive diagnostic tools for assessing heart health. It records the electrical activity of the heart and provides valuable

information about rhythm, conduction, and timing. ECG waveforms consist of distinct components such as the P wave, QRS complex, and T wave, each representing a specific phase of the cardiac cycle. Deviations or irregularities in these waveforms may indicate abnormal heart function or an elevated risk of myocardial infarction.

Despite its clinical importance, ECG analysis largely depends on manual interpretation by trained medical professionals. This process requires expertise and careful evaluation, and interpretations may vary between clinicians. Moreover, long-duration ECG recordings generate large volumes of data, making manual analysis both challenging and time-consuming. In emergency scenarios, even minor delays in interpretation can significantly affect treatment outcomes. Additionally, short-term ECG recordings may fail to capture intermittent or early-stage abnormalities, increasing the risk of missed diagnoses.

To address these limitations, automated ECG analysis systems have been introduced. However, many existing approaches rely on predefined rules and threshold-based methods that lack adaptability to natural variations in ECG patterns caused by age, physiological differences, and health conditions. These systems are also vulnerable to noise resulting from patient movement, improper electrode placement, and external electrical interference. With the rapid adoption of wearable devices and home-based monitoring technologies, there is a growing demand for intelligent systems capable of analyzing ECG data accurately, continuously, and in real time.

Artificial Intelligence (AI) has emerged as a powerful solution for enhancing ECG interpretation. Machine learning algorithms can analyze large datasets, identify complex patterns, and detect subtle abnormalities that may not be easily recognized through conventional methods. AI-based systems enable automated feature extraction, efficient signal processing, and accurate classification of cardiac conditions. By integrating preprocessing techniques, feature extraction,

and machine learning models, such systems can deliver faster and more reliable diagnostic support while reducing the workload on healthcare professionals.

This paper presents the development of an AI-based heart attack detection system using ECG signal analysis. The proposed system is designed to facilitate early identification of abnormal cardiac activity and support continuous monitoring. It can operate with ECG data obtained from wearable or portable devices, making it suitable for hospitals, home care, and remote healthcare environments. By enabling timely alerts and accurate analysis, the proposed approach contributes to improved preventive cardiac care, reduced emergency risks, and enhanced overall heart health management.

II. LITERATURE REVIEW

This paper evaluates various real-time ECG preprocessing techniques within an edge-cloud computing framework, focusing on energy efficiency, computational complexity, and real-time performance. The study finds that adaptive filtering methods are more suitable than traditional fixed bandpass filters for handling time-varying ECG signals. Optimized Pan-Tompkins and ICA+LDA approaches are identified as efficient and accurate solutions for QRS detection, while artifact detection layers and signal quality-aware techniques offer strong performance for signal quality assessment on resource-constrained edge devices. Overall, the results support the use of lightweight, adaptive preprocessing methods to enable accurate, energy-efficient, and privacy-preserving real-time ECG analysis[1].

This paper presents a transformer-based approach, specifically using the Vision Transformer (ViT), for automated ECG-based heart disease detection by overcoming the limitations of traditional CNN and RNN models in capturing long-range dependencies. By leveraging self-attention mechanisms, the proposed model effectively extracts both local and global patterns from ECG images, improving classification performance and

generalization. Through preprocessing, data augmentation, and optimized training strategies, the model achieves high expected accuracy (around 80–90%) and demonstrates robustness across multiple heart condition classes. Comparative analysis shows that while RNNs and CNNs capture temporal and local features respectively, Vision Transformers provide a more comprehensive representation of ECG signals. Overall, the study highlights the strong potential of transformer-based models for scalable, accurate, and automated cardiac diagnosis in real-world healthcare applications[2].

This paper presents an AI-based approach for ECG signal classification using deep learning techniques to improve automated heart disease detection. The study focuses on effective preprocessing, feature extraction, and model optimization to enhance classification accuracy and robustness. By leveraging advanced neural network architectures, the proposed system captures important ECG patterns and achieves reliable performance across different cardiac conditions. Overall, the research highlights the potential of intelligent, data-driven models for accurate, efficient, and scalable ECG-based cardiac diagnosis in modern healthcare systems[3].

This paper proposes an Intelligent Decision Support System for cardiac arrhythmia management by integrating AI algorithms with ECG data to enhance detection and classification accuracy. The system applies preprocessing and feature extraction techniques, followed by a hybrid ensemble of CNN, LSTM, and SVM models to enable real-time and reliable arrhythmia prediction. Compared to traditional rule-based approaches, the proposed method demonstrates higher accuracy, sensitivity, and computational efficiency while adapting continuously to evolving arrhythmia patterns. Overall, the study highlights the potential of AI-driven ECG fusion systems to improve diagnostic precision, support clinical decision-making,

and enhance patient outcomes in cardiac care[4].

This paper presents a deep learning-based framework for automated ECG signal classification aimed at improving early detection of cardiovascular diseases. The proposed system incorporates effective preprocessing and feature extraction techniques to enhance signal quality before applying optimized neural network models for accurate classification. Experimental results demonstrate improved performance in terms of accuracy, sensitivity, and reliability compared to conventional approaches. Overall, the study emphasizes the significance of intelligent, data-driven ECG analysis systems for enabling efficient, scalable, and clinically supportive cardiac diagnosis[5].

This paper presents a smart edgecomputing wearable device for real-time, beat-by-beat ECG classification using an ANN-based AI model integrated directly into a low-power ARM Cortex-based SoC. The system performs on-device preprocessing, feature extraction, and classification to detect five types of ECG beats with an overall accuracy of 90.6%, while maintaining low power consumption (maximum 128 mW) and low latency (6 ms). Trained on the MITBIH dataset and validated through both test data and real-time trials on human subjects, the wearable demonstrates reliable and efficient cardiac monitoring. Overall, the study highlights the potential of AI-enabled, low-power wearable systems for continuous, real-time detection of cardiac abnormalities[6].

III. RELATED WORK

Recent advancements in artificial intelligence and signal processing have significantly improved electrocardiogram (ECG) analysis for early detection of cardiac abnormalities. Freudenberg et al. (2024) emphasized the importance of real-time preprocessing techniques to remove noise, baseline drift, and motion artifacts, thereby improving signal quality for accurate diagnosis

[2] Ganai et al. (2024) proposed a Transformer-based deep learning model capable of capturing temporal dependencies in ECG signals, leading to higher classification accuracy than traditional methods [3]. Additionally, Rahman et al. (2024) highlighted the growing use of wearable devices for continuous cardiac monitoring, enabling early diagnosis and supporting preventive healthcare [1].

Further research has focused on enhancing real-time detection and predictive performance. Morshed et al. (2023) developed an edge-AI framework for real-time ECG classification on portable devices, reducing latency and improving responsiveness during emergencies. Dhingra et al. (2023) demonstrated through comparative analysis that ensemble machine learning algorithms provide greater robustness and accuracy than individual classifiers for heart disease prediction [4]. Similarly, Khan et al. (2023) showed that deep neural networks can effectively identify complex ECG patterns, reducing reliance on manual interpretation while improving diagnostic consistency [5]. Panindre et al. (2022) concluded that AI-based monitoring systems can generate early warnings for myocardial infarction, thereby helping to reduce mortality associated with delayed detection. Overall, these studies indicate that integrating preprocessing methods, deep learning techniques, and wearable technologies enhances diagnostic reliability and supports continuous cardiac monitoring, although challenges such as computational complexity and real-time implementation remain.

IV. System Architecture of AI-Based Heart Attack Detection Using ECG Signal Analysis

The proposed system utilizes artificial intelligence techniques to detect potential heart attacks through ECG signal analysis. The methodology consists of multiple stages, including data acquisition, signal preprocessing, feature extraction, machine learning-based classification, and automated report generation.

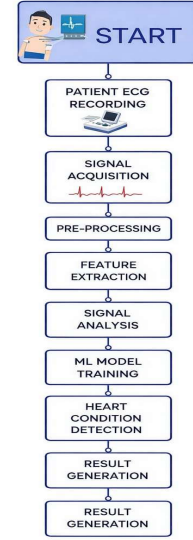


Fig. 1. System Architecture of AI-Based Heart Attack Detection Using ECG Signal Analysis

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Fig. 1 illustrates the overall architecture of the proposed AI-based heart attack detection system. The process begins with ECG data acquisition, followed by signal preprocessing to remove noise and improve signal quality. Important features are then extracted and provided to the machine learning model for classification. Based on the prediction results, the system generates alerts and reports to support early diagnosis and continuous heart monitoring.

A. Data Acquisition

The first step in the proposed system involves collecting ECG signals that represent the electrical activity of the heart. Electrodes are placed on the patient's body to capture these signals, which are then transmitted to an AD8232 sensor module. The sensor amplifies weak cardiac signals and filters out initial noise to ensure reliable signal capture.

The AD8232 module is interfaced with an Arduino microcontroller, which converts the analog ECG signals into digital form for computational analysis. The acquired data is continuously transmitted to a computer system, enabling real-time monitoring of heart

activity. Accurate data acquisition is essential because the performance of the entire detection system depends on the quality of the recorded ECG signals.



Fig. 2. Signal Aquisition

B. Signal Preprocessing

Raw ECG signals often contain unwanted disturbances caused by muscle contractions, patient movement, electrode misplacement, and external electrical interference. Such noise can distort waveform patterns and negatively impact classification accuracy.

To address this issue, preprocessing techniques are applied to enhance signal quality. Filtering methods are used to remove highfrequency and low-frequency noise, while baseline correction eliminates signal drift. Normalization is then performed to standardize the signal amplitude, making the data suitable for further analysis.

Effective preprocessing ensures that only meaningful cardiac information is retained, thereby improving the reliability and robustness of the detection system.

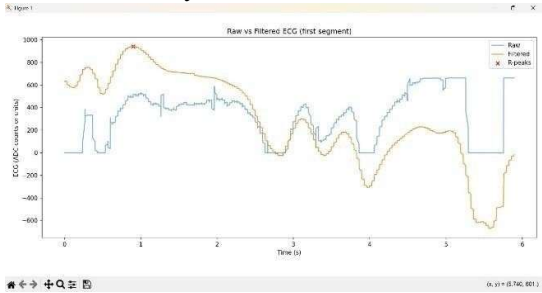


Fig. 3.Signal Preprocessing

C. Feature Extraction

Feature extraction is a crucial step in the proposed system as it converts the cleaned ECG signal into meaningful information that can be used for accurate classification. After preprocessing, the ECG waveform is analyzed to identify important characteristics that represent the electrical activity of the heart. Each heartbeat contains distinct components such as the P wave, QRS complex, and T wave, which provide valuable insights into cardiac behavior.

In this work, significant features such as heart rate, R–R interval, PR interval, QRS duration, and signal amplitude are extracted from the ECG data. These parameters help differentiate between normal and abnormal heart activity. Detecting variations in these features allows the system to recognize patterns associated with potential cardiac abnormalities.

By transforming raw ECG signals into structured numerical data, feature extraction reduces data complexity while preserving clinically important information. The extracted features are then forwarded to the machine learning model, enabling faster processing and improving the overall accuracy of heart attack detection.

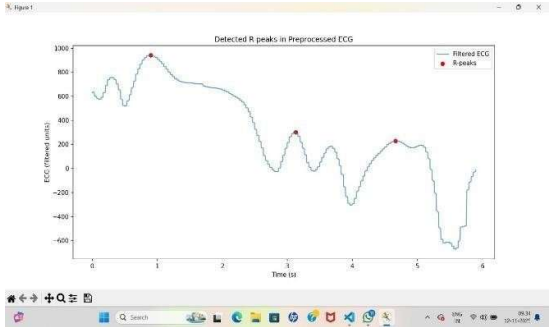


Fig .4.Feature Extraction

D. Machine Learning Model Training and Classification

The machine learning stage plays a crucial role in automatically identifying cardiac abnormalities from extracted ECG features. After preprocessing and feature extraction, the dataset containing statistical and physiological parameters is used to train the classification model. The features and labels are first separated, where the feature set represents ECG

characteristics and the labels indicate whether the signal corresponds to a normal condition or a heart attack. To prepare the data for model training, categorical labels are encoded into numerical form to ensure compatibility with machine learning algorithms. The dataset is then divided into training and testing subsets so that the model can learn patterns from known data and be evaluated on unseen samples.

A Random Forest classifier is employed to perform the classification task due to its robustness and ability to handle complex feature relationships. The model is trained on the prepared dataset and later tested to measure its performance using evaluation metrics such as accuracy, confusion matrix, and classification report.

Once satisfactory performance is achieved, the trained model is saved for future predictions. For any new ECG sample, the system extracts the required features, loads the saved model, and predicts whether the condition is normal or indicative of a heart attack. Finally, the prediction results are displayed and stored for further analysis.

E. Prediction and Report Generation

The final stage of the proposed system focuses on generating predictions and presenting the analysis results in a clear and meaningful format. After the trained machine learning model classifies the ECG signal, the system determines whether the heart activity is normal or indicates a potential cardiac risk. For each processed ECG sample, the extracted features are provided as input to the saved classification model, which produces the prediction output. Based on the result, the system displays the detected condition to the user, enabling quick understanding of the heart status. This automated prediction mechanism reduces the time required for manual interpretation and supports faster decisionmaking.

In addition to real-time prediction, the system stores the results for future reference and analysis. Maintaining these records helps in tracking changes in heart activity over time and

can assist healthcare professionals in identifying patterns that may require medical attention. The report generated by the system improves accessibility of diagnostic information and strengthens continuous heart health monitoring.

V. Implementation

Algorithm: ECG Signal Preprocessing and R-Peak Detection

Step 1: Start.

Step 2: Check if *ecg_data.csv* exists.

If not, display error and stop.

Step 3: Read the CSV file and verify it contains Time and ECG_Value columns.

If missing, display error and stop.

Step 4: Convert the Time column into datetime format and compute the median time difference (dt).

Step 5: Calculate sampling frequency:

$$fs = 1 / dt$$

Step 6: Extract ECG values into array *raw* and apply filters:

- High-pass filter (0.5 Hz)
- 50 Hz notch filter
- Bandpass filter (0.5–40 Hz, order 4)

Step 7: Save the filtered signal into *ecg_preprocessed.csv*.

Step 8: Set peak detection parameters:

$$\text{Minimum distance} = 0.25 \times fs$$

$$\text{Prominence} = 0.5 \times \text{standard deviation of filtered signal}$$

Step 9: Detect R-peaks using peak detection method.

Step 10: Save peak index and peak time into *ecg_peaks.csv*.

Step 11: If number of peaks ≥ 2 , compute RR intervals and calculate:
Heart Rate = $60 / \text{RR interval}$
Find mean and median heart rate.

Step 12: Convert time into seconds relative to start, plot raw signal, filtered signal, and detected R-peaks.

step 13: Stop.

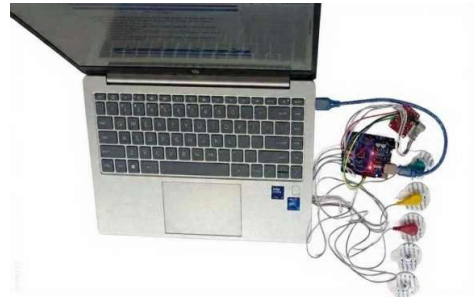
Hardware Implementation:

- Components Used:
- AD8232 ECG Sensor Module
- Arduino (Microcontroller)
- ECG Electrodes and Connecting Wires
- USB Cable
- Laptop with Arduino IDE and Python Environment



Hardware Setup:

1. ECG electrodes were placed on the chest at standard positions.
2. Electrodes were connected to the AD8232 ECG module.
3. The sensor module was connected to Arduino analog input pins.
4. Arduino was connected to the laptop via USB cable.
5. Arduino IDE Serial Plotter was used to visualize the real-time ECG wave-form.
6. This setup enabled real-time capturing of biological ECG signals.



• Software Implementation:

Preprocessing of ECG Signals:

The raw ECG signal contained various types of noise and fluctuations that could affect the accuracy of subsequent analysis. To improve the clarity of the signal, a Python preprocessing script was used to perform several essential steps. These included removing baseline drift, filtering out high-frequency noise, normalizing the signal amplitude, eliminating sudden spikes, and detecting R-peaks. Together, these preprocessing steps ensured that the ECG data became clean, stable, and suitable for accurate feature extraction and model training. The cleaned signal was saved as *ecgpreprocessed.csv*.

Feature Extraction:

Features were extracted in two ways:

a) Feature Extraction for One Sample

- R-peak detection using `findpeaks()`
- Heart Rate (BPM)
- RR Interval Mean and Standard Deviation
- Signal Mean, Standard Deviation, Minimum, Maximum

b) Feature Extraction for All Samples

For each CSV file, the following features were extracted:

- Mean
- Standard Deviation
- Maximum
- Minimum

- Skewness
- Kurtosis
- RMS (Root Mean Square)
- Label (Normal / Heart Attack) Machine

Learning Model Implementation:

Model Used:

A Random Forest Classifier was used for ECG classification.

Input: Extracted ECG Features Output: Normal / Heart Attack Prediction

V. Results and Discussion

The proposed AI-based heart attack detection system was evaluated to determine its effectiveness in analyzing ECG signals and identifying abnormal cardiac activity. After preprocessing and feature extraction, the machine learning model was tested using prepared ECG data to verify its classification



emphasizing the need for timely medical attention.

These results confirm that the proposed system can successfully differentiate between multiple heart conditions and support early risk identification. Therefore, the developed model serves as an effective tool for automated cardiac monitoring and assists in improving diagnostic efficiency.



Fig. 6 Software Output Showing Normal Condition



Fig. 7 Software Output Showing Mild-Risk Condition

Fig. 8 Software Output Showing High-Risk Detection

capability. The system successfully distinguished between normal and abnormal heart conditions, demonstrating its suitability for automated cardiac monitoring.

The software interface clearly presents prediction results along with automated reports, allowing quick interpretation of cardiac conditions. Fig. 6 illustrates the system output for a normal heart condition, indicating stable cardiac activity. Fig. 7 shows the detection of a moderate-risk condition, where the system identifies potential irregularities that may require medical observation. Fig. 8 displays the high-risk prediction generated by the system, highlighting critical abnormalities and

Furthermore, the proposed approach demonstrates the practical application of artificial intelligence in healthcare. By combining signal analysis with machine learning, the system provides a supportive tool for early diagnosis and preventive care. Although the system is designed for academic implementation, its framework can be extended for real-time monitoring environments such as hospitals, home-based care, and wearable health devices.

Overall, the results indicate that the developed system has the potential to improve

early heart risk detection, enhance diagnostic support, and contribute to more effective heart health management.

VI. Conclusion

This paper presented the design and development of an AI-based heart attack detection system using ECG signal analysis. The proposed approach integrates preprocessing, feature extraction, and machine learning techniques to automatically identify abnormal heart activity. By enabling accurate classification of cardiac conditions, the system supports early detection and reduces reliance on manual ECG interpretation.

The developed model demonstrates the potential of artificial intelligence in improving healthcare diagnostics by providing faster and more reliable analysis. Continuous monitoring combined with automated prediction can assist healthcare professionals in making timely decisions and may help lower the risk of severe cardiac events.

Overall, the proposed system contributes to preventive healthcare by enhancing heart health monitoring and promoting early medical intervention. With further advancements, such intelligent systems can play a significant role in modern clinical and remote healthcare environments.

VII. Future Scope

The AI-based ECG heart attack detection system can be further enhanced by integrating it with wearable devices for continuous heart monitoring and early detection of abnormalities. Future improvements may include cloud-based analysis for efficient data storage and sharing with healthcare professionals, as well as edge AI for real-time detection with better privacy. The addition of multiple biosensors such as SpO₂ and blood pressure could improve prediction accuracy, making the system a reliable tool for preventive cardiac care.

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